

Laboratory Exercise

Wireless Attacks

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Task 1: Wi-Fi Packet Analysis using Wireshark

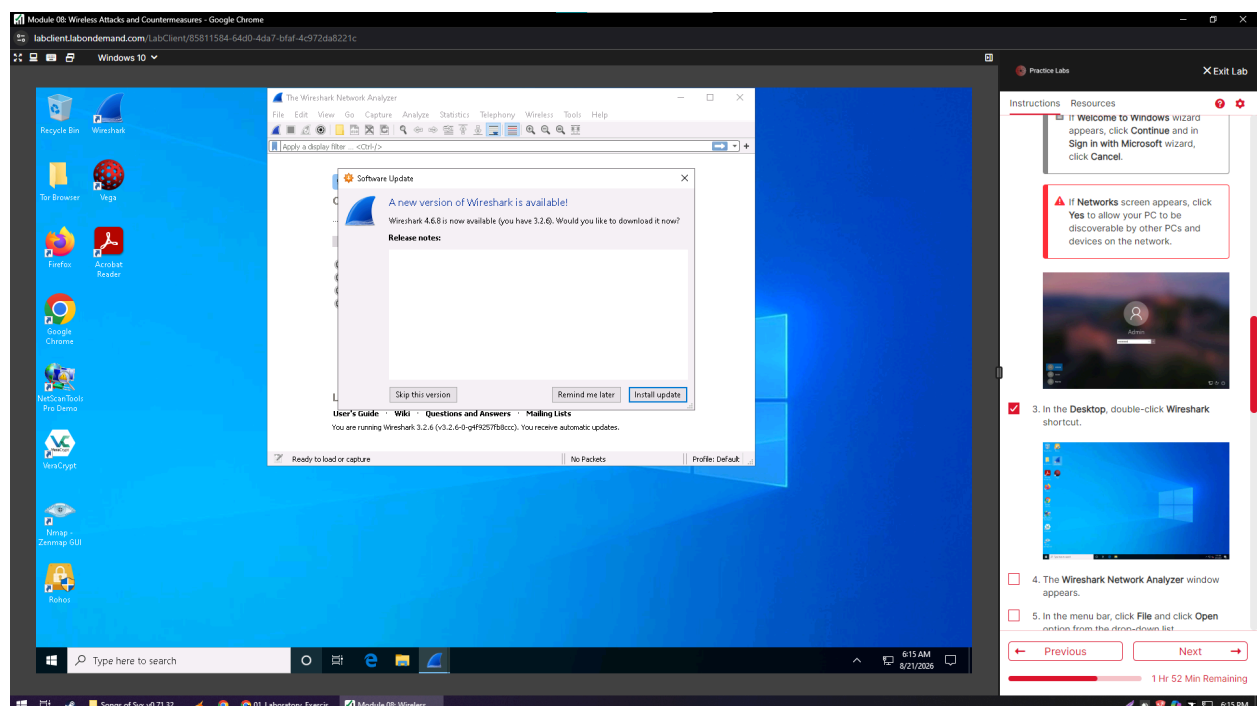
1. By default, the **Windows 10** machine selected, click **Ctrl+Alt+Delete**.
Alternatively, you can also click **Ctrl+Alt+Delete** button under **Windows 10** machine thumbnail in the **Resources** pane or Click **Ctrl+Alt+Delete** button under Commands (**thunder** icon) menu.
2. By default, the **Admin** user profile is selected, click **Pa\$\$w0rd** or paste the password in the **Password** field and press **Enter** to login.

Notes:

If the **Welcome to Windows** wizard appears, click **Continue** and in **Sign in with Microsoft** wizard, click **Cancel**.

If the **Networks** screen appears, click **Yes** to allow your PC to be discoverable by other PCs and devices on the network.

3. In the **Desktop**, double-click the **Wireshark** shortcut.
4. The **Wireshark Network Analyzer** window appears.



5. In the menu bar, click **File** and click **Open** option from the drop-down list.
6. **Wireshark: Open Capture File** window appears, navigate to **D:\EHE-Tools\EHE Module 08 Wireless Attacks and Countermeasures\Sample Captures**, select **WEPcrack-01.cap**, and click **Open**.

The screenshot displays a Windows 10 desktop environment. The primary window is the 'Wireshark - Open Capture File' dialog box. It shows a file explorer view of the 'Sample Captures' directory. A table lists the following files:

Name	Date modified	Type
WEPcrack-01.cap	1/13/2018 4:00 PM	Wireshark
WEPcrack-01.csv	1/13/2018 4:00 PM	Microphone
WEPcrack-01.kismet.netml	1/13/2018 4:00 PM	Microphone
WPA2crack-01.cap	1/16/2018 1:55 PM	Wireshark
WPA2crack-01.csv	1/16/2018 1:55 PM	Microphone
WPA2crack-01.kismet.netml	1/16/2018 1:55 PM	Microphone

The 'WEPcrack-01.cap' file is selected. The 'File name' field contains 'WEPcrack-01.cap' and 'Files of type' is set to 'All Files'. The 'Open' button is highlighted. Below the file list, the 'Read filter' is set to 'Automatically detect file type' and the 'Format' is 'Wireshark/pcapng'. The 'Size' is 195 MB and the 'Start / elapsed' time is 2018-01-13 04:36:10 / 00:04:11.

In the background, the Wireshark main window is visible. The menu bar shows 'File' > 'Open'. To the right, a 'Practice Labs' sidebar is open, showing a list of instructions. The first five instructions are checked, and the sixth instruction, which matches the current step, is unchecked. The sidebar also shows a progress bar at the bottom indicating '1 Hr 49 Min Remaining'.

6. **Wireshark: Open Capture File** window appears, navigate to **D:\EHE-Tools\EHE Module 08 Wireless Attacks and Countermeasures\Sample Captures**, select **WEPcrack-01.cap**, and click **Open**.
7. The **WEPcrack-01.cap** file opens in **Wireshark** window, showing you the details of the packet for analysis. Here you can see the wireless packets captured which were otherwise masked to look like **ethernet** traffic. Provide a screenshot of this.

Here 802.11 protocol indicates wireless packets.

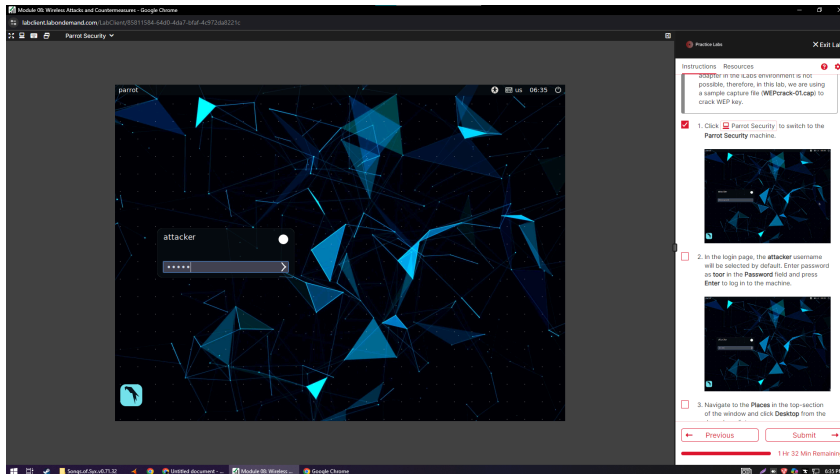
You can access the saved packet capture file anytime, and by issuing packet filtering commands in the **Filter** field, you can narrow down the packet search in an attempt to find packets containing sensible information.

In real time, attackers enforce packet capture and packet filtering techniques to capture packets containing passwords (only for websites implemented on the HTTP channel), perform attacks such as session hijacking, and so on.

The screenshot displays a Windows 10 desktop environment. The main window is Wireshark, which has opened the file 'WEPcrack-01.cap'. The packet list on the left shows a series of 802.11 Beacon frames from 'Netgear_e4:38:100' to 'Apple_c9:b2:11c (78...)'. The packet details pane on the right shows the structure of an IEEE 802.11 Beacon frame, including fields like 'Frame Control', 'Duration', 'Address', 'Sequence Number', and 'Data'. The packet bytes pane at the bottom shows the raw hex and ASCII data of the selected packet. On the far right, a 'Practice Labs' sidebar is visible, containing instructions for the lab. The instructions are numbered 3 through 6, with step 6 currently selected and highlighted in red. Step 6 reads: '6. Wireshark: Open Capture File window appears, navigate to D:\EHE-Tools\EHE Module 08 Wireless Attacks and Countermeasures\Sample Captures, select WEPcrack-01.cap and click Open.' The sidebar also includes a 'Previous' button, a 'Next' button, and a progress bar indicating '1 Hr 36 Min Remaining'.

Lab 2 – Task 1: Crack a WEP Network using Aircrack-ng

1. Click on **Parrot Security** to switch to the Parrot Security machine.
2. In the login page, the **attacker** username will be selected by default. Enter the password as **toor** in the **Password** field and press **Enter** to log in to the machine.

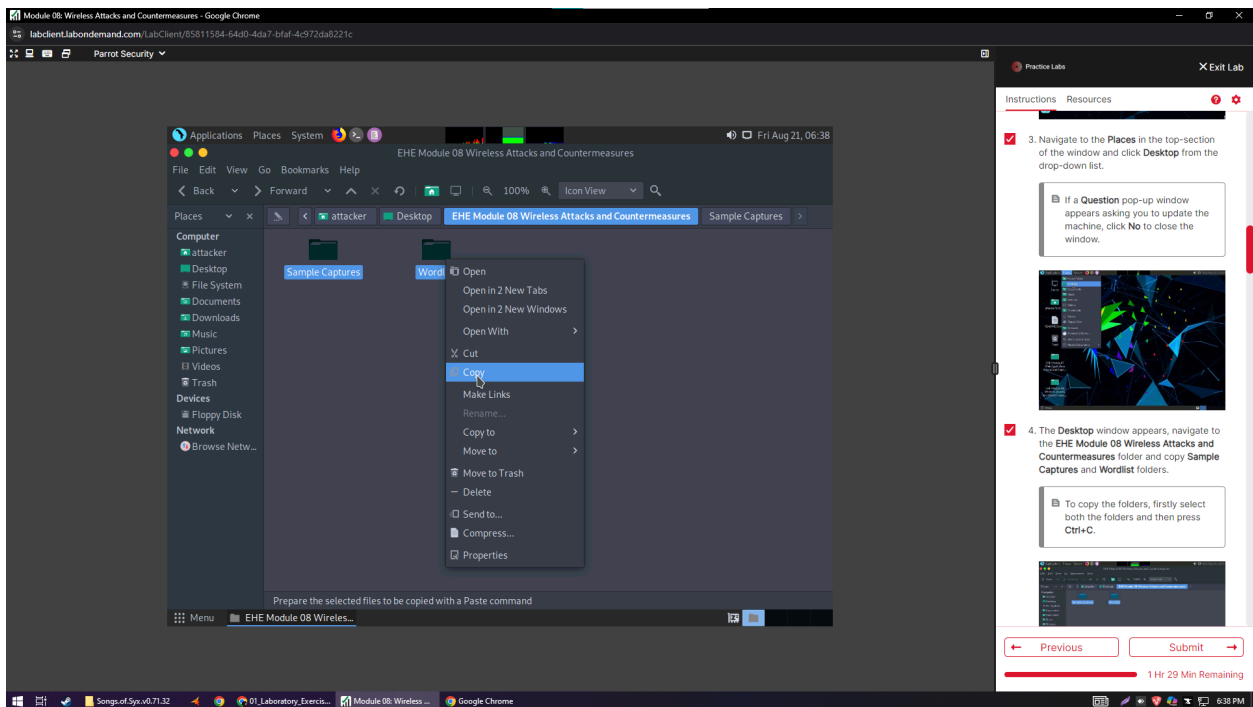


3. Navigate to the **Places** in the top section of the window and click **Desktop** from the drop-down list.

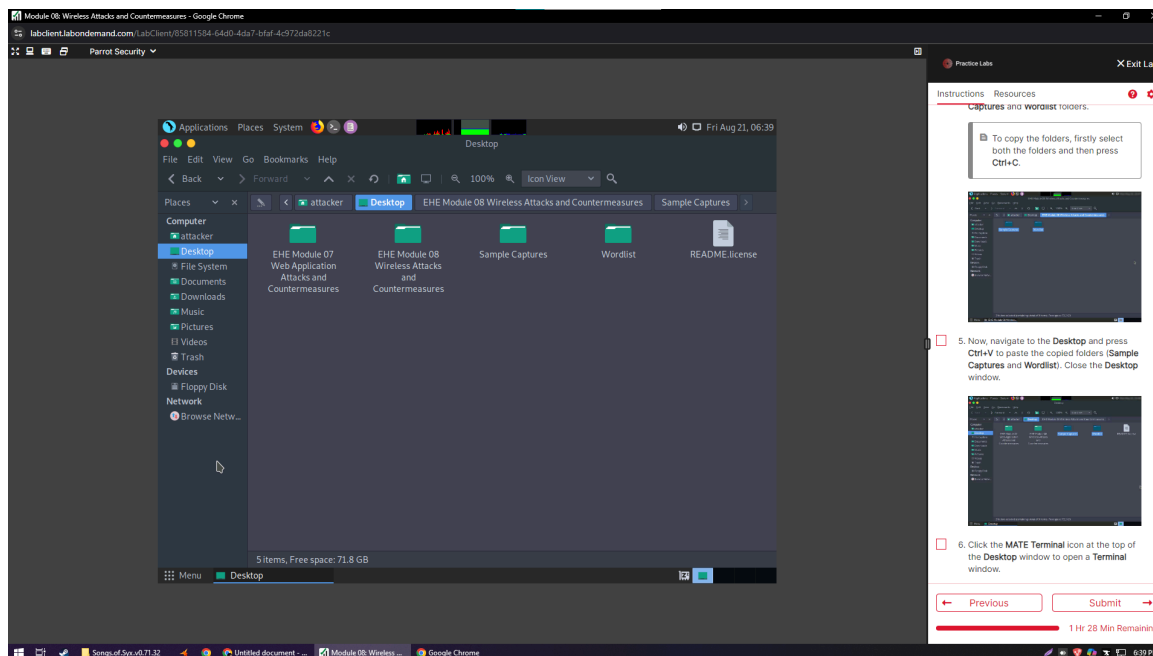
Note:

*If a **Question** pop-up window appears asking you to update the machine, click **No** to close the window.*

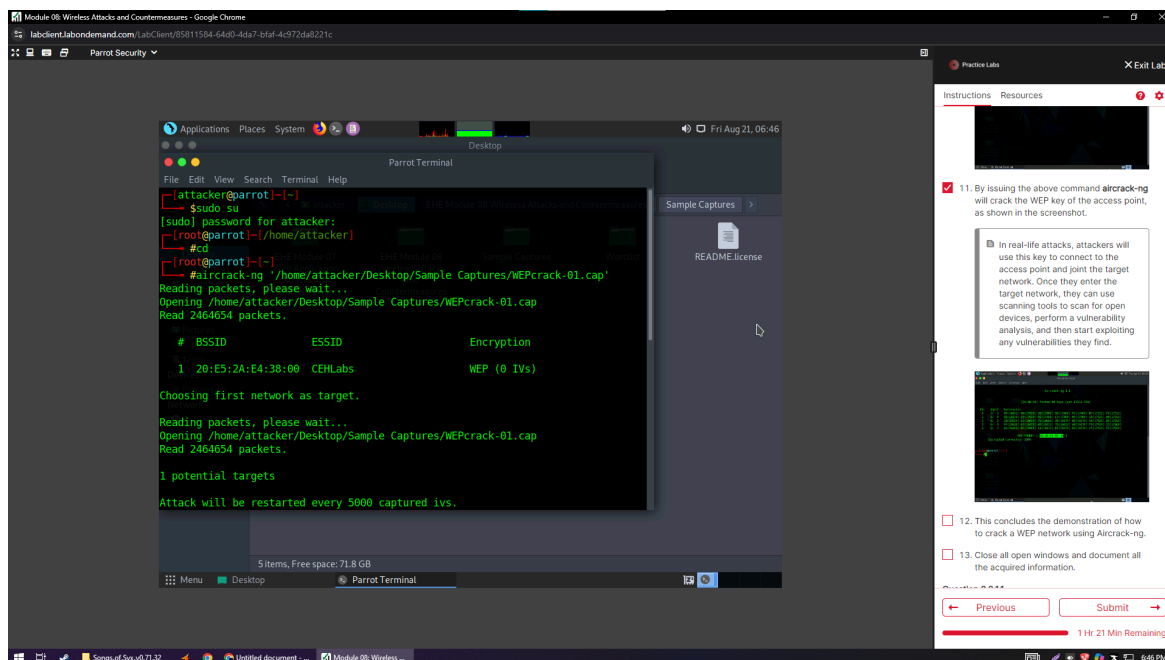
4. The **Desktop** window appears, navigate to the **EHE Module 08 Wireless Attacks and Countermeasures** folder, and copy the **Sample Captures** and **Wordlist** folders.



- Now, navigate to the **Desktop** and press **Ctrl+V** to paste the copied folders (**Sample Captures** and **Wordlist**). Close the **Desktop** window.



- Click the **MATE Terminal** icon at the top of the **Desktop** window to open a **Terminal** window.
- Parrot Terminal** window appears. In the terminal window, type **sudo su** and press **Enter** to run the programs as a root user.
- In the **[sudo] password for attacker** field, type **toor** as a password and press **Enter**.
- Now, type **cd** and press **Enter** to jump to the root directory.



10. In the **Parrot Terminal** window, type **aircrack-ng /home/attacker/Desktop/Sample Captures/WEPCrack-01.cap** and press **Enter**. Provide the screenshot of the result.
11. By issuing the above command, **aircrack-ng** will crack the WEP key of the access point.
12. Use the Aircrack-ng suite on the Parrot Security machine to crack the WEP encryption of a Wi-Fi network. Screenshot the key found in this exercise.

Note: Sample captured Wi-Fi packets are available in the EHE Module 08 Wireless Attacks and Countermeasures\Sample Captures folder on the desktop.

The screenshot shows a Parrot Security virtual machine environment. The main window is a Parrot Terminal running the **aircrack-ng** command. The terminal output shows the process of testing 88 keys and finding a WEP key. The output is as follows:

```
[00:00:00] Tested 88 keys (got 13614 IVs)
KB  depth  byte(vote)
0  2/ 3    98(18432) 88(17920) 38(17408) 50(17408) FC(17408)
1  3/ 8    48(18176) 33(17920) 92(17408) C3(17408) 05(17408)
2  0/ 2    31(20224) 15(18688) 76(18688) 38(18176) 8C(18176)
3  0/ 1    97(22016) 03(19456) 48(18432) 7D(18432) AB(18176)
4  0/ 2    49(20480) BF(19968) 14(18432) D7(18176) E8(18176)

KEY FOUND! [ 98:48:35:97:49 ]
Decrypted correctly: 100%
```

The terminal prompt is **root@parrot:~#**. The sidebar on the right contains a "Practice Labs" section with instructions and a "Submit" button. The instructions include:

- 11. By issuing the above command **aircrack-ng** will crack the WEP key of the access point, as shown in the screenshot.
- 12. This concludes the demonstration of how to crack a WEP network using Aircrack-ng.
- 13. Close all open windows and document all the acquired information.

The sidebar also features a "Previous" button, a "Submit" button, and a progress bar indicating "1 Hr 20 Min Remaining".